# SpaceX and Starship

SPACE TRANSPORTATION | Aiming for full reuse to change space-transport costs

## What is SpaceX?

A US space company developing and operating rockets and spacecraft. It also provides the Starlink satellite communications service.

## What is Starship?

A super-heavy space-transport system comprising the upper-stage spacecraft Starship and the first-stage Super Heavy booster. It aims to operate with both stages reused.

## The concept: size, reuse and frequent operation

Carrying large payloads at once and using vehicles repeatedly aims to reduce space-transport costs substantially. Beyond satellite launches, it envisages transport to the Moon and Mars. Distinguish the full-reuse concept from stable commercial operations.

# What would successful implementation change?

HOW IT WORKS | Greater capacity and frequency expand options for space use

## Satellite communications | Many satellites into orbit

Efficiently launching many satellites could facilitate building and renewing networks, potentially increasing communications capacity.

## Moon and Mars | Moving people and supplies

The concept includes personnel and supplies for lunar activities and future Mars transport. Long-term operations still face technical, safety and other challenges.

## Space industry | More equipment and applications

Moving large equipment and supplies could broaden space activities such as research, observation and manufacturing.

## The key: repeated operation

Transport costs depend on reuse counts, maintenance costs and launch frequency. Beyond successful flights, the efficiency of sustained operations determines commercial viability.

# Connections with Japanese companies

JAPANESE COMPANIES | Distinguish SpaceX business ties from broader space-industry involvement

### KDDI (9433) | Starlink communications services

Offers “au Starlink Direct” and has signed a service agreement with SpaceX for next-generation “Starlink Mobile V2” satellites. This is a communications link, distinct from supplying Starship parts.

### Toray (3402) | Carbon-fiber composites for space

Supplies materials for rockets, satellites and other uses. It is involved in lightweight space equipment, but uses and adopted materials depend on the vehicle. This document has not confirmed direct supply to Starship.

### Mitsubishi Electric (6503) | Satellites and space equipment

Works in satellites, onboard equipment and ground facilities. It has links to expanding space use, but this does not establish direct Starship dealings or effects on orders and profits.

### NEC (6701) | Satellites, ground facilities and data

Works in satellites, ground facilities and data processing/analysis. It has links to expanding space use, but this does not establish direct Starship dealings or effects on orders and profits.

# Space-market growth and corporate earnings

INDUSTRY STRUCTURE | More transport does not automatically increase each company's profit

## Communications | Adoption and profitability

Wider satellite-communications use may expand service coverage and applications. Effects on operators' earnings depend on user numbers, contract terms, equipment costs and operating costs.

## Materials | Each vehicle uses different materials

More rocket and satellite production may increase material demand. Effects depend on designs, adopted materials and supply agreements. Starship's success alone does not determine any company's profit growth.

## Space equipment | Demand growth and competition

Demand for satellites, ground facilities and data use may grow. Some fields may also face competition from SpaceX and others. Market growth and individual companies' earnings are separate issues.

Starship aims to transport large amounts of supplies to space repeatedly. If lower transport costs are achieved, possibilities for space use expand.